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An Explainable and Optimized Machine Learning Framework for Multisensor Remote Sensing-Based Land Use and Land Cover Classification

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ABSTRACT

Land use and land cover (LULC) mapping supports environmental monitoring and spatial planning, but classification performance often suffers in feature spaces that combine multiple data sources with redundant predictors. This study compares five feature selection methods (mRMR, ReliefF, Boruta, RFE, HSIC-Lasso) with three tree-based classifiers (RF, XGBoost, LightGBM) for LULC classification in Kahramankazan, Ankara, Türkiye. Combining Sentinel-1, Sentinel-2, topographic, climatic, and socioeconomic data yielded 240 candidate features. Hyperparameters were tuned with Optuna under spatial cross-validation, and SHAP was used to interpret feature contributions. HSIC-Lasso + LightGBM achieved the highest F1 (0.9655) and OA (0.9657) using only 31 features, 84% fewer than Boruta + RF, which reached a comparable F1 with 196 features. McNemar's test with Holm–Bonferroni correction confirmed equivalence with the best Boruta variants and superiority over the mRMR- and ReliefF+RF baselines. HSIC-Lasso also completed the pipeline 16.3 times faster than Boruta, supporting compact embedded selection in multi-source LULC workflows.

1 | Introduction

Anthropogenic pressures have substantially reshaped Earth's land surfaces over recent decades; global land-use change affected nearly one-third of the terrestrial surface between 1960 and 2019 (Winkler et al. 2021), and humans currently use or manage approximately three-quarters of Earth's ice-free land area (Meyfroidt et al. 2022). Urban expansion drives extensive and often persistent transformations of terrestrial ecosystems (Seto et al. 2011), agriculturally driven habitat loss threatens global biodiversity (Tilman et al. 2017), and multiple anthropogenic stressors degrade freshwater systems (Vörösmarty et al. 2010); these processes are underpinned by demographic growth and shifting consumption patterns (Díaz et al. 2019). The resulting degradation, including soil erosion, deforestation, habitat fragmentation, and biodiversity loss, extends well beyond local boundaries and threatens human well-being (Foley

et al. 2005; Hooke et al. 2012). Land-use change also markedly affects the functioning of key biophysical components, including atmospheric composition (Heald and Spracklen 2015), the global carbon cycle (Mahowald et al. 2017), and the hydrological cycle (Bosmans et al. 2017). Land use and land cover (LULC) mapping enables consistent characterization of surface features over large areas (Gómez et al. 2016; Phiri et al. 2020) and facilitates analyses of ecosystem service provision (Hasan et al. 2020), hydrological regulation (Sterling et al. 2013), and land surface temperature dynamics (Mohamed et al. 2025).

Satellite-based remote sensing (RS) provides the observational foundation for such monitoring, delivering consistent, repeat-pass observations of Earth's surface across broad spatial extents and multi-decadal time spans (Fu et al. 2024; Wulder et al. 2019). The Landsat program, launched in 1972, has since been joined by the Sentinel constellations, and

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together they provide optical observations at 10–30 m resolution (Phiri et al. 2020). Despite these advantages, optical sensors are constrained by cloud cover, weather, and illumination. Furthermore, in heterogeneous urban and peri-urban settings, classes such as buildings, roads, vegetated surfaces, and bare land share similar spectral signatures and cannot be reliably separated using optical data alone (Haas and Ban 2017; Zhang et al. 2018). In contrast, synthetic aperture radar (SAR) systems, operating in the microwave region, overcome the environmental limitations by acquiring data independent of weather and illumination, while also capturing surface roughness, moisture, and structural properties (Irfan et al. 2025; Moreira et al. 2013). The joint use of SAR and optical data has been shown to improve land cover classification, with overall accuracy gains reported for integrated Sentinel-1/Sentinel-2 classifications in heterogeneous landscapes (Clerici et al. 2017), per-class improvements documented for built-up and bare soil classes in a multi-class urban scheme (Tavares et al. 2019), and gains for separating built-up areas from barren land, categories that share spectral signatures in optical imagery (Shrestha et al. 2021). Nevertheless, the magnitude of fusion-driven accuracy gains varies across landscapes; auxiliary and ancillary predictors are therefore commonly added to enhance class separability. Beyond the primary satellite reflectance and backscatter bands, two additional categories of predictors are commonly distinguished: ancillary inputs derived from the satellite data themselves, including spectral indices, SAR-derived indices, and gray-level co-occurrence matrix (GLCM) textural features; and auxiliary inputs from external sources, such as topographic parameters derived from digital elevation models (DEMs), climatic layers, and socioeconomic proxies like nightlight data and population density (Kavzoglu and Bilucan 2023; Venter and Sydenham 2021). The benefits of multi-source integration are well-established in heterogeneous landscapes, with documented gains from optical-auxiliary fusion (Hurskainen et al. 2019), and from the addition of SAR data (Amoakoh et al. 2021; Ibrahim 2023). Integrating such diverse sources generates high-dimensional and heterogeneous datasets, which exceed the capacity of traditional parametric classifiers (e.g., maximum likelihood) that assume specific data distributions. Machine learning (ML) techniques address this by modeling nonlinear relationships among large numbers of predictors (Lary et al. 2016; Sheykhoumoussa et al. 2020). Random forest (RF) has been widely adopted in RS for its robustness against overfitting and its capacity to process high-dimensional data efficiently (Belgiu and Drăguț 2016). Gradient boosting methods such as XGBoost and LightGBM have shown competitive or superior performance to RF in specific contexts, including Sentinel-2-based land cover classification (Colkesen and Ozturk 2022) and SAR-optical fusion for built-up area mapping (Shao et al. 2024). However, their relative advantage in multi-source Sentinel-1/Sentinel-2 workflows over heterogeneous peri-urban landscapes remains underexamined, particularly in combination with feature selection methods. Although RF and boosted decision trees are relatively robust to high-dimensional feature spaces given sufficient training data (Maxwell et al. 2018), classifier performance can degrade as dimensionality grows and redundant or noisy features accumulate (Ewees et al. 2025). Feature selection and hyperparameter optimization are therefore critical to the reliability of ML classification (Abdi 2020; Georganos et al. 2018).

A systematic comparative analysis was therefore conducted in the Kahramankazan district of Ankara, Türkiye. The study integrated Sentinel-2 optical imagery and Sentinel-1 SAR data acquired between 1 June and 30 September 2024. From these data, satellite-derived feature layers were generated, encompassing Sentinel-2 reflectance bands, spectral indices, Sentinel-1 SAR backscatter and derived indices (from both ascending (ASC) and descending (DSC) orbits), and gray-level co-occurrence matrix (GLCM) textural features. These were supplemented with auxiliary layers including topographic features from a digital elevation model (DEM), climatic features, and socioeconomic proxies such as population density and nightlight data. Five feature selection methods, namely minimum redundancy maximum relevance (mRMR), Boruta, ReliefF, Hilbert-Schmidt independence criterion Lasso (HSIC-Lasso), and recursive feature elimination (RFE), were evaluated in combination with three ML algorithms (RF, XGBoost, and LightGBM), with hyperparameters optimized via Optuna-based search under spatial cross-validation. The resulting models were assessed on an independent test set, and feature importance for the best-performing model was analyzed using SHapley Additive exPlanations (SHAP). To the best of our knowledge, the pairing of five feature selection methods with three tree-based classifiers within a multi-source Sentinel workflow has not previously been reported.

2 | Material and Method

2.1 | Study Area

Kahramankazan district, situated in the northwestern part of Ankara Province, Türkiye, was selected as the study area (Figure 1). The district covers approximately 581 km² at elevations of 800–1700 m, forming a plateau-like morphology typical of Central Anatolia. Its climate is semi-arid continental, with hot, high-evaporation summers and cold, dry winters. Kahramankazan is a peri-urban landscape dominated by cultivated areas but also containing forests, urban surfaces, bare land, and water bodies. This heterogeneity in reflectance, topography, and surface roughness exceeds what a single data source can reliably capture, making the district a suitable test site for multi-source classification.

2.2 | Dataset Composition

The dataset integrates five data sources (optical, SAR, topographic, climatic, and socioeconomic) to represent both the spectral and structural characteristics of land cover. In this study, “multi-source” refers to this diversity of data sources rather than to temporal diversity. Optical and SAR observations were acquired over the summer months of 2024, while topographic, climatic, and socioeconomic layers were obtained from long-term or annual data products. Following the categorization outlined in the Introduction, the features fall into three groups: primary inputs comprising the Sentinel-2 reflectance bands and Sentinel-1 raw backscatter bands (14 features); ancillary inputs derived from these primary data, including spectral indices, SAR-derived indices, and GLCM textural features (215 features); and auxiliary inputs from external sources, including topographic, climatic, and socioeconomic layers (11 features); 240 features in total.

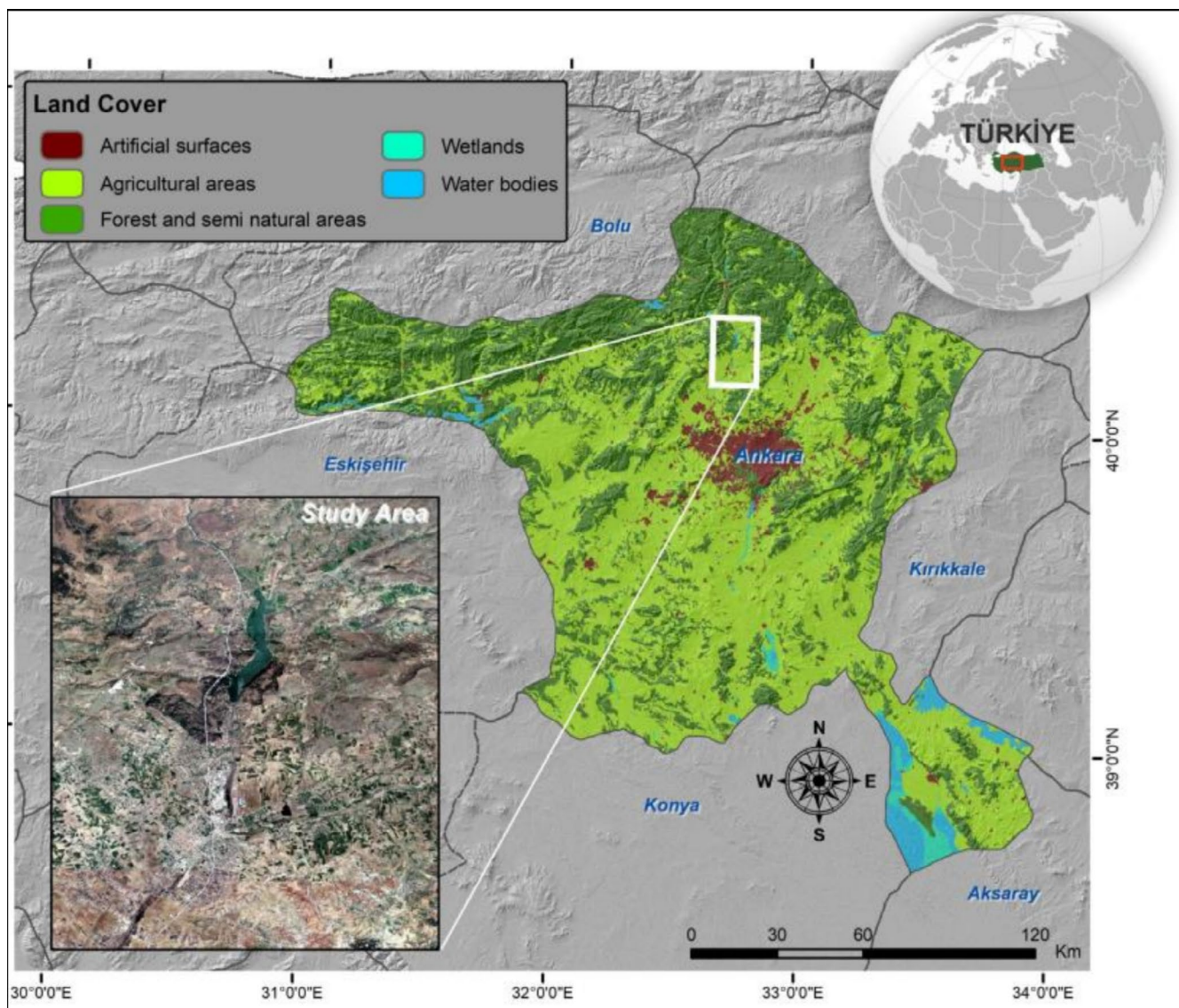


FIGURE 1 | Geographical location of the study area in Kahramankazan district, northwestern Ankara, Türkiye.

2.2.1 | Sentinel Data

The analysis relies on two primary satellite data sources: Sentinel-1 SAR and Sentinel-2 optical imagery. Both were accessed and processed through the Google Earth Engine (GEE) cloud computing platform. The optical data consist of Sentinel-2 Level-2A surface reflectance products obtained from the *COPERNICUS/S2_SR_HARMONIZED* collection, covering the period from 1 June to 30 September 2024. Ten optical bands were used in the analysis: B2 (Blue), B3 (Green), B4 (Red), B5-B7 (Red edge 1–3), B8 (NIR), B8A (Narrow NIR), B11 (SWIR1), and B12 (SWIR2). The remaining bands (B1 for aerosol retrieval and B9 for water vapor) were excluded, as they are intended for atmospheric correction rather than surface characterization. From these 10 bands, 13 spectral indices were derived as ancillary features to represent water, forest, urban, bare land, road, fallowland, and cultivated classes. To complement the optical data, Sentinel-1 SAR imagery was obtained from the *COPERNICUS/S1_GRD_FLOAT* collection, which provides the backscatter coefficient (σ^0) in linear power. The full preprocessing chain, including border noise removal, radiometric terrain correction to gamma naught (γ^0), speckle filtering, and conversion to

decibels, is described in Section 2.3.1. Vertical-Vertical (VV) and Vertical-Horizontal (VH) polarizations were retained for both ASC and DSC orbits as primary backscatter features in dB. Three additional ancillary indices were then derived for each orbit from the γ^0 backscatter in linear power: the VV/VH polarization ratio, the radar vegetation index (RVI), and the normalized difference polarization index (NDPI). This produced five SAR features per orbit: two primary backscatter bands and three ancillary derived indices. The mathematical formulations and references of the spectral and SAR indices are provided in Table 1.

2.2.2 | Auxiliary Data

To complement the Sentinel data, topographic, climatic, and socioeconomic auxiliary features were integrated into the dataset (Table 2). Topographic information was derived from the Advanced Land Observing Satellite Phased Array type L-band Synthetic Aperture Radar (ALOS PALSAR) Radiometrically Terrain Corrected DEM obtained from the Alaska Satellite Facility, with an original spatial resolution of 12.5m, which

was resampled to 10 m using nearest-neighbor interpolation to ensure spatial compatibility with Sentinel-2 imagery. From this DEM, five fundamental topographic features were generated: elevation, slope, aspect, topographic position index (TPI), and terrain ruggedness index (TRI). TPI and TRI were computed following the formulations of Weiss (2001) and Riley et al. (1999), respectively. In addition to terrain characteristics, four climatic features (mean summer temperature, cumulative summer precipitation, soil moisture, and water deficit) were obtained from the TerraClimate dataset (Abatzoglou et al. 2018), covering the period from 1 June to 30 September 2024 to align with the Sentinel acquisition window. Finally, WorldPop 2024 population density and Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB) nightlight data were used as socioeconomic indicators to capture human activity and urban intensity (Bondarenko et al. 2025; Elvidge et al. 2017). All auxiliary layers with coarser native resolutions (TerraClimate at ~4 km, VIIRS at ~500 m, WorldPop at ~100 m) were resampled

to 10 m using nearest-neighbor interpolation within GEE to match the Sentinel-2 grid. This method preserves the original pixel values without introducing interpolation artifacts, although the effective information content of each layer remains at its native resolution.

2.2.3 | GLCM Textural Features

GLCM textural features were integrated to capture local spatial patterns. GLCM models the spatial dependency structure of surfaces by considering the probabilities of co-occurring gray levels of neighboring pixel pairs at a given distance and orientation. In this study, the 14 fundamental GLCM textural features introduced by Haralick et al. (1973) were employed: angular second moment (ASM), contrast (CON), correlation (COR), variance (VAR), homogeneity (HOM), sum average (SA), sum variance (SVAR), sum entropy (SENT), entropy (ENT), difference

TABLE 1 | Optical and SAR indices with corresponding formulations and references.

Type	Index	Formula	References
Optical	Normalized Difference Vegetation Index (NDVI)	$\frac{B8 - B4}{B8 + B4}$	Rouse et al. (1974)
Optical	Enhanced Vegetation Index (EVI)	$2.5 \frac{B8 - B4}{B8 + 6B4 - 7.5B2 + 1}$	Huete et al. (2002)
Optical	Soil-Adjusted Vegetation Index (SAVI)	$\frac{B8 - B4}{B8 + B4 + L} (1 + L)$	Huete (1988)
Optical	Green-Red Vegetation Index (GRVI)	$\frac{B3 - B4}{B3 + B4}$	Motohka et al. (2010)
Optical	Modified Normalized Difference Water Index (MNDWI)	$\frac{B3 - B11}{B3 + B11}$	Xu (2006)
Optical	Land Surface Water Index (LSWI)	$\frac{B8 - B11}{B8 + B11}$	Xiao et al. (2005)
Optical	Normalized Difference Built-up Index (NDBI)	$\frac{B11 - B8}{B11 + B8}$	Zha et al. (2003)
Optical	Urban Index (UI)	$\frac{B12 - B8}{B12 + B8}$	Kawamura et al. (1996)
Optical	Bare Soil Index (BSI)	$\frac{(B11 + B4) - (B8 + B2)}{(B11 + B4) + (B8 + B2)}$	Rikimaru et al. (2002)
Optical	Dry Bare-Soil Index (DBSI)	$\frac{B11 - B3}{B11 + B3} - \frac{B8 - B4}{B8 + B4}$	Rasul et al. (2018)
Optical	Index-Based Built-up Index (IBI)	$\frac{NDBI - (MNDWI + SAVI) / 2}{NDBI + (MNDWI + SAVI) / 2}$	Xu (2008)
Optical	Built-up Area Extraction Index (BAEI)	$\frac{B4 + L}{B3 + B11}$	Bouzekri et al. (2015)
Optical	Normalized Difference Tillage Index (NDTI)	$\frac{B11 - B12}{B11 + B12}$	van Deventer et al. (1997)
SAR	Radar Vegetation Index (RVI)	$\frac{4\gamma_{VH}^0}{\gamma_{VV}^0 + \gamma_{VH}^0}$	Nasirzadehdizaji et al. (2019)
SAR	Normalized Difference Polarization Index (NDPI)	$\frac{\gamma_{VV}^0 - \gamma_{VH}^0}{\gamma_{VV}^0 + \gamma_{VH}^0}$	Cao et al. (2008)

Note: In Table 1, L is the soil-brightness correction factor ($L = 0.5$ for SAVI and 0.3 for BAEI, following the respective original references).

TABLE 2 | Summary of auxiliary datasets, variable types, and feature counts used in this study.

Category	Data source	Variables	Feature count
Topographic	ALOS PALSAR DEM	Elevation, Slope, Aspect, TPI, TRI	5
Climatic	TerraClimate	Mean Summer Temperature, Cumulative Summer Precipitation, Soil Moisture, Water Deficit	4
Socioeconomic	WorldPop 2024	Population Density	1
	VIIRS DNB Monthly	Nightlight Intensity	1

TABLE 3 | Mathematical definitions and functional descriptions of the 14 GLCM textural features (Hall-Beyer 2017; Haralick et al. 1973).

Feature	Mathematical definition	Description
ASM	$ASM = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} p(i,j)^2$	Measures textural uniformity; higher values indicate more homogeneous regions
CON	$CON = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} (i-j)^2 p(i,j)$	Quantifies local intensity variations and edge strength
COR	$COR = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \frac{(i-\mu_x)(j-\mu_y)p(i,j)}{\sigma_x \sigma_y}$	Indicates linear dependency of gray levels in neighboring pixels
VAR	$VAR = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} p(i,j)(i-\mu_x)^2$	Dispersion of gray levels around the marginal mean
HOM	$HOM = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \frac{p(i,j)}{1+(i-j)^2}$	Measures local uniformity of neighboring gray levels
SA	$SA = \sum_{k=0}^{2n-2} k p_{x+y}(k)$	Mean of the sum distribution
SENT	$SENT = - \sum_{k=0}^{2n-2} p_{x+y}(k) \ln(p_{x+y}(k))$	Measures randomness in the sum distribution; higher values of SENT denote a greater degree of disorder
SVAR	$SVAR = \sum_{k=0}^{2n-2} p_{x+y}(k)(k-SA)^2$	Dispersion of the sum distribution around its mean
ENT	$ENT = - \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} p(i,j) \ln(p(i,j))$	Randomness of gray-level co-occurrences; higher values indicate irregular textures
DENT	$DENT = - \sum_{k=0}^{n-1} p_{x-y}(k) \ln(p_{x-y}(k))$	Measures randomness (uncertainty) in the gray-level difference distribution
DVAR	$DVAR = \sum_{k=0}^{n-1} p_{x-y}(k)(k-\mu_{x-y})^2$	Variability of gray-level differences around their mean
IMC1	$HX = - \sum_i p_x(i) \ln(p_x(i))$ $HY = - \sum_j p_y(j) \ln(p_y(j))$ $HXY1 = - \sum_{i,j} p(i,j) \ln(p_x(i)p_y(j))$ $IMC1 = \frac{ENT - HXY1}{\max(HX, HY)}$	Mutual dependence between the GLCM marginals, normalized by the larger marginal entropy
IMC2	$HXY2 = - \sum_{i,j} p_x(i)p_y(j) \ln(p_x(i)p_y(j))$ $IMC2 = \sqrt{1 - \exp[-2(HXY2 - ENT)]}$	Deviation from gray-level independence; bounded between zero and one
MCC	$Q(i,j) = \sum_{k=0}^{n-1} \frac{p(i,k)p(j,k)}{p_x(i)p_y(j)}$ $MCC = \sqrt{\lambda_2}$, where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ are the eigenvalues of Q	Higher-order gray-level dependencies beyond linear correlation; from the second eigenvalue of the auxiliary matrix

variance (DVAR), difference entropy (DENT), information measure of correlation I (IMC1), information measure of correlation II (IMC2), and maximal correlation coefficient (MCC) (Table 3). These metrics were computed in Python using a 3×3 window at a distance of 1 pixel across four directional offsets (0° , 45° , 90° , and 135°), with values averaged across directions for rotational invariance. Gray levels were quantized to 64 levels using equal-probability (rank-based) quantization. At a 10m spatial resolution, this window size covers approximately 30×30 m, capturing local texture variation while preserving detail at class boundaries; larger windows risk mixing textures across adjacent land cover classes in heterogeneous peri-urban areas, where class patches are often smaller than 50×50 m. These parameters were applied to 10 optical bands from Sentinel-2 and four SAR polarization layers from Sentinel-1 (VV and VH in both ASC and DSC orbits, in dB), producing 14 textural features per band and 196 GLCM textural features in total.

2.3 | Methods

This study applies an eight-stage workflow for land cover classification based on multi-source RS data and ML algorithms. The workflow comprises the following stages (Figure 2): (i) preprocessing of Sentinel-2 optical and Sentinel-1 SAR imagery (Section 2.3.1); (ii) computation of spectral and SAR indices (Section 2.2.1); (iii) extraction of GLCM textural features (Section 2.2.3); (iv) generation of training and test datasets through a spatial split strategy (Section 2.3.2); (v) application of five feature selection methods (Section 2.3.3); (vi) classification with three ML algorithms (RF, XGBoost, and LightGBM) and Optuna-based hyperparameter optimization (Sections 2.3.4–2.3.5); (vii) feature importance analysis using SHAP (Section 2.3.6); and (viii) performance evaluation, statistical significance testing, and class separability analysis (Section 2.3.7).

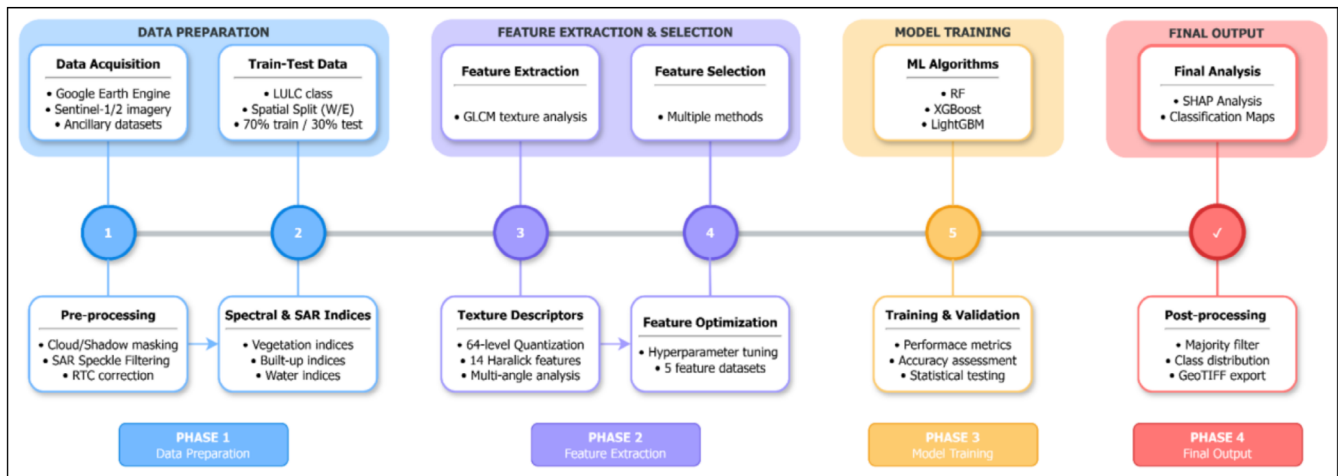


FIGURE 2 | Methodological workflow of the study.

2.3.1 | Preprocessing

To construct the dataset defined in Section 2.2, the raw satellite imagery was preprocessed as follows. For Sentinel-2 optical data, scenes with more than 10% cloud cover were excluded, and clouds, shadows, and cirrus were masked using the QA60 quality band and the Scene Classification Layer. A median composite was then generated from the remaining scenes. For Sentinel-1 SAR data, images were retrieved as σ^0 values, which are processed by GEE with thermal noise removal, radiometric calibration, and geometric terrain correction using the Shuttle Radar Topography Mission (SRTM) 30m DEM by default. ASC and DSC orbits were processed separately to preserve geometric diversity, and additional preprocessing steps were applied to each orbit. First, border noise was removed using thresholds on incidence angle and backscatter values. Radiometric terrain correction was then performed using the ALOS PALSAR DEM (12.5m native resolution, resampled to 10m) to convert σ^0 to γ^0 (normalized to the local incidence angle), compensating for topographic effects on backscatter. Layover and shadow areas were subsequently masked based on local incidence angle and slope geometry. Speckle noise was then reduced using an adapted dual-window Lee filter (Lee 1980) with 3×3 and 7×7 kernels, using the coefficient of variation to distinguish homogeneous, heterogeneous, and point-target regimes following the thresholds of Lopes et al. (1990). Finally, a temporal median composite was produced for each orbit and converted to decibels.

2.3.2 | Creation of Training and Test Samples

Following the preprocessing stage, training and test datasets based on seven land cover classes were prepared: water (lakes, rivers, reservoirs), forest (dense natural or plantation tree cover), urban (residential, commercial, and industrial structures), bare land (non-vegetated sandy, rocky, or soil surfaces), road (asphalt or stabilized transport networks), fallowland (temporarily uncultivated agricultural fields), and cultivated (actively cropped land). Class labels were assigned through visual interpretation of high-resolution imagery in Google Earth acquired in 2024 contemporaneous with the Sentinel observation window, a common approach for LULC sample generation (Gündüz 2025;

Li et al. 2020; Nasiri et al. 2022). Because fallowland and bare land show comparable spectral signatures within a summer composite, multi-temporal Google Earth imagery outside the June–September window was also examined: fallowland parcels showed vegetation cover in other seasons and displayed regular, rectangular boundaries adjacent to cultivated parcels, whereas bare land sites lacked vegetation across all available dates and had irregular boundaries on naturally exposed soil or slopes. At 10m resolution, roads were also difficult to delineate: minor roads are narrower than the pixel footprint, and asphalt surfaces share reflectance properties with both bare land and urban surfaces. Road was included as a separate class because the primary roads in the study area are wider than 10m and are reliably captured at the Sentinel-2 resolution. Beyond these labeling considerations, a spatial split strategy was applied to mitigate spatial autocorrelation, dividing the study area along a north–south-oriented boundary into an eastern training region and a western test region (Figure 3), thereby reducing spatial dependence between training and test samples. The two regions were comparable in class distribution: all seven classes were naturally present in both subsets, and class-distribution similarity was statistically confirmed (Cramér's $V=0.152$; Jensen–Shannon distance=0.132). On top of this natural similarity, samples were drawn equally from each class within each region, resulting in identical class composition in the training and test sets. A balanced dataset was constructed with a 70:30 training–test split, producing 4900 training samples (700 per class) and 2100 test samples (300 per class). In addition, the training samples were further partitioned into five spatial folds via k -means clustering on the sample coordinates ($k=5$), which served as the grouping variable in a GroupKFold scheme. This ensured that samples belonging to the same spatial cluster were never split between training and validation partitions, preserving spatial independence in the cross-validation procedure. This scheme was then applied during both feature selection (Section 2.3.3) and hyperparameter optimization (Section 2.3.5).

2.3.3 | Feature Selection Methods

As the inclusion of all 240 features in the dataset poses a risk of overfitting, feature selection was applied to reduce noise and improve

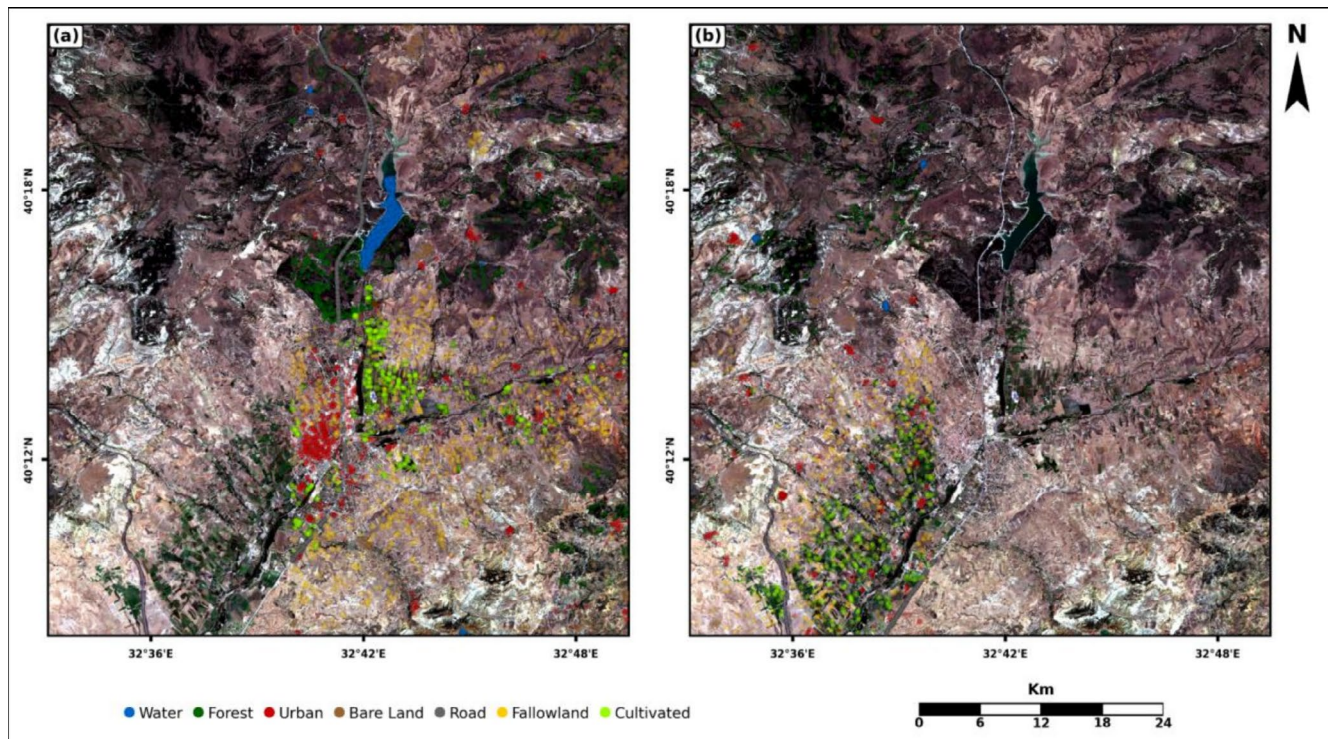


FIGURE 3 | Training (a) and test (b) sample locations used for LULC classification.

classification accuracy. Five algorithms were selected to represent the three main categories of feature selection: filter-based methods (ReliefF and mRMR), wrapper-based methods (Boruta and RFE), and an embedded method (HSIC-Lasso). These categories differ in how they integrate the classifier into the selection process: filter methods evaluate features independently of any classifier, wrapper methods use iterative classifier feedback, and embedded methods perform selection within model fitting. Since no single category has been shown to be uniformly optimal for fused optical-SAR-auxiliary feature spaces, all three were evaluated on the same dataset to examine which approach best addresses the redundancy among features in multi-source LULC inputs.

The *ReliefF* algorithm was applied based on similarity relationships between each sample and its nearest neighbors, evaluating within-class similarity and between-class separability simultaneously. The method assigns feature weights according to how well each feature distinguishes neighboring samples belonging to the same or different classes (Kononenko 1994). Since ReliefF produces feature importance scores but does not determine the optimal subset size, correlation-based feature selection (CFS) merit analysis (Hall 1999) was incorporated to guide the selection of the subset size, with the merit-scanning bounds tuned jointly with the ReliefF parameters in the Optuna framework (Section 2.3.5).

The *mRMR* algorithm was applied to reduce redundant information among features. The method evaluates each feature's relevance to the class labels alongside its similarity to other features; features with high relevance but low redundancy are retained (Peng et al. 2005).

The *Boruta* algorithm was used to perform feature-importance analysis based on the RF classifier. Shadow features were

generated by shuffling the values of each original feature. Each feature was then compared against the maximum importance score of the shadow features, and statistical significance was assessed. Based on this comparison, features were classified as confirmed, rejected, or tentative (Kursa and Rudnicki 2010).

The *RFE* algorithm iteratively removed features that contributed minimally to the model. At each iteration, the feature with the lowest contribution to classification performance was identified and eliminated; this process continued until the pre-defined number of features was reached (Guyon et al. 2002).

The *HSIC-Lasso* algorithm was used to quantify feature-label dependencies through the underlying HSIC measure. The method operates within a kernel-based framework, which allows the assessment of both linear and nonlinear relationships among features from different data sources, such as optical, SAR, and topographic layers. Feature selection was performed by solving a Lasso optimization problem on the resulting dependency measures (Yamada et al. 2014). In this study, the block HSIC-Lasso variant was adopted to improve scalability for the high-dimensional feature space. This variant divides the samples into blocks to reduce memory requirements (Climente-González et al. 2019).

2.3.4 | Machine Learning

Three ML algorithms were used to classify the feature subsets obtained after feature selection: RF, XGBoost, and LightGBM. These algorithms were selected because they perform robustly on high-dimensional and heterogeneous datasets and are widely used in LULC classification studies (Colkesen and Ozturk 2022; Hussain et al. 2025; McCarty et al. 2020).

RF is an ensemble learning method composed of many decision trees, each trained on a bootstrapped sample with a random subset of features considered at each split. The final classification is determined by majority voting across all trees, which reduces the risk of overfitting (Breiman 2001).

XGBoost is a gradient-boosted decision tree algorithm that combines computational efficiency with high predictive accuracy. It optimizes the loss function with regularization terms, which help prevent overfitting, and achieves strong performance on datasets with complex data dependencies (Chen and Guestrin 2016).

LightGBM is a gradient-boosting method that uses histogram-based decision trees and a leaf-wise tree-growth strategy. By binning continuous feature values into discrete histograms, it achieves faster training and lower memory consumption than standard gradient-boosting implementations (Ke et al. 2017; Ustuner and Sanli 2019).

2.3.5 | Hyperparameter Optimization

Hyperparameter optimization was applied using the Optuna framework for systematic exploration of the search space (Akiba et al. 2019). Optimization was conducted in two stages: first, the hyperparameters of each feature selection method were tuned; second, the hyperparameters of the three ML algorithms were optimized independently for each resulting feature subset. For the feature selection methods, the tuned parameters were the number of nearest neighbors (k), distance-weighting bandwidth (σ), and lower (min_features) and upper (max_features) bounds of the CFS merit search for ReliefF; the number of selected features (K), relevance score (F -statistic (f), Kolmogorov–Smirnov (KS), or Random Forest importance (rf)), and aggregation function (denominator) for mRMR (with the redundancy measure fixed at Pearson correlation); the maximum tree depth (max_depth), maximum iterations (max_iter), p -value threshold (α), and percentile cut-off (percentile) for Boruta; the number of estimators (n_estimators), max_depth, step size (step), and features to retain (n_to_select) for RFE; and the number of selected features

(num_feat), block size (B), permutations (M), and neighbors (max_neighbors) for the block HSIC-Lasso. For the three ML algorithms, n_estimators and max_depth were tuned for all three; min_samples_split and min_samples_leaf for RF; and learning_rate and subsample for XGBoost and LightGBM. The full search ranges are listed in Tables 5 and 6. Each optimization was run for 100 trials under the cross-validation scheme described in Section 2.3.2, using F1 as the evaluation metric. A separate optimization was performed for each of the 15 combinations of feature selection method and ML algorithm, and the best hyperparameter configuration from each search was used to train the final models for evaluation on the independent test dataset.

2.3.6 | SHapley Additive exPlanations

SHAP analysis was applied to improve the interpretability of the model that achieved the highest F1 on the independent test set. SHAP quantifies the marginal contribution of each feature to the prediction based on game-theoretic attribution values (Lundberg and Lee 2017). Given the tree-based structure of the classifiers, TreeSHAP was applied as a computationally efficient variant of SHAP designed for tree ensemble models (Lundberg et al. 2020). A class-stacked SHAP summary bar plot was generated using mean absolute SHAP values, enabling both the identification of globally important features and the visualization of class-specific contributions across the seven land cover classes.

2.3.7 | Model Evaluation and Class Separability

Classification performance was assessed using confusion matrix-based accuracy metrics commonly used in the RS literature (Biscoveanu et al. 2025; Gündüz 2026; Krivoguz et al. 2023): OA, precision (PCC), recall (RC), F1, false alarm rate (FA), missed detection rate (MD), and specificity (SP) (Table 4). For the multi-class problem, F1, PCC, RC, FA, MD, and SP were computed using a one-vs-rest scheme, where each class in turn was treated as the positive class against the remaining six, yielding true positive (TP), false positive (FP), false negative (FN), and true negative (TN) counts per class. The resulting per-class

TABLE 4 | Accuracy metrics and statistical test for classification performance assessment.

Accuracy metric	Description	Formula
OA	Proportion of correctly classified samples over the total number of samples	$OA = \frac{TP + TN}{TP + FP + TN + FN}$
PCC	Proportion of correctly predicted positive samples among all predicted positives	$PCC = \frac{TP}{TP + FP}$
RC	Proportion of correctly predicted positive samples among all actual positives	$RC = \frac{TP}{TP + FN}$
F1	Harmonic mean of precision and recall	$F1 = \frac{2PCCRC}{PCC + RC}$
FA	Proportion of false positives among all negative samples	$FA = \frac{FP}{TN + FP}$
MD	Proportion of false negatives among all positive samples	$MD = \frac{FN}{TP + FN}$
SP	Proportion of correctly identified negative samples	$SP = \frac{TN}{TN + FP}$
McNemar's test (χ^2)	Statistical test for assessing the significance of differences between two classifiers	$\chi^2 = \frac{(n_{ij} - n_{ji} - 1)^2}{n_{ij} + n_{ji}}$

Note: For multi-class classification, OA is equivalently computed as $OA = \sum_{i=1}^C TP_i / N_{total}$, where TP_i is the count of correctly classified samples in class i , C is the number of classes ($C = 7$), and N_{total} is the total number of test samples.

scores were then averaged across the seven classes using arithmetic mean (macro-averaging), giving equal weight to each class regardless of class size. In addition, to test whether performance differences between classifier configurations were statistically significant, McNemar's test (McNemar 1947) was applied. This non-parametric test evaluates the symmetry of error patterns between two classifiers applied to the same test set, using a chi-square statistic with Edwards' continuity correction (Edwards 1948). Since pairwise comparisons across the 15 configurations resulted in 105 tests, the Holm-Bonferroni procedure was used to adjust for multiple testing (Holm 1979), and significance was evaluated on the adjusted p -values at $\alpha=0.05$.

To assess class separability in the multi-source feature space, class-conditional feature distributions were compared using mean $\pm$ standard deviation and Cohen's d as the effect-size measure (Cohen 1988), with values of 0.2, 0.5, and 0.8 conventionally interpreted as small, medium, and large effects. The comparison was performed pairwise across all feature groups (SAR backscatter, topography, climate, socioeconomic layers, GLCM textures, and spectral indices) to identify which sources separate which class pairs.

3 | Results

Five feature selection methods (ReliefF, mRMR, Boruta, RFE, and HSIC-Lasso) were tuned with Optuna to identify discriminative features from the 240 candidate features, and the optimized hyperparameters are summarized in Table 5.

Among the filter-based methods, ReliefF was tuned with CFS merit analysis as described in Section 2.3.3, with the merit-scanning window treated as an additional pair of Optuna hyperparameters (min_features, max_features) to balance merit-based redundancy control with classification performance. Optuna converged to a window of [66, 125], within which the merit reached its maximum at 66 features (merit=0.443, cross-validation F1=0.931). A separate diagnostic analysis over the unrestricted range [2, 150] revealed an unconstrained merit peak at 26 features (merit=0.595, Figure 4); this smaller subset, however, yielded lower cross-validation performance, supporting the broader subset selected by Optuna (F1-driven), with CFS merit determining the in-window subset size.

The distributions of selected features across spectral, textural, and auxiliary categories are summarized in Figure 5. The number of selected features varied substantially across algorithms: HSIC-Lasso produced the most compact subset (31 features), followed by RFE (39), ReliefF and mRMR (66 each), and Boruta (196 of 240). Differences across methods were most pronounced for the auxiliary and ancillary feature groups. For topographic features, Boruta and mRMR included all five (elevation, slope, aspect, TPI, and TRI), whereas HSIC-Lasso preserved only elevation and TRI, RFE kept only elevation, and ReliefF excluded all topographic features. Climatic features were fully retained by Boruta, RFE, and mRMR (temperature, precipitation, soil moisture, and water deficit), while HSIC-Lasso preserved only soil moisture and water deficit, and ReliefF excluded all climatic features. For socioeconomic features, ReliefF and Boruta included both population and nightlight, while HSIC-Lasso and

TABLE 5 | Optimal hyperparameters for feature selection methods.

Method	Parameter	Optimized value
Boruta	max_depth	10
	max_iter	198
	α	0.002
	percentile	99
HSIC-Lasso	num_feat	31
	B	59
	M	4
	max_neighbors	18
mRMR	K	66
	relevance	KS
	denominator	Mean
ReliefF	k	32
	σ	55
	min_features	66
	max_features	125
RFE	n_estimators	497
	max_depth	11
	n_to_select	39
	step	3

Note: Search ranges: Boruta: max_depth [7–15], max_iter [100–400], α [0.001–0.01], percentile [85–100]; HSIC-Lasso: num_feat [20–150], B [20–60], M [2–10], max_neighbors [5–20]; mRMR: K [20–150], relevance [f , ks , rf], denominator {mean, max}; ReliefF: k [10–100], σ [15–60], min_features [20–75], max_features [20–150]; RFE: n_estimators [100–500], max_depth [8–15], n_to_select [20–150], step [1–5].

mRMR retained only population, and RFE selected only nightlight. Within the textural category, Boruta included the broadest set, followed by ReliefF and mRMR; HSIC-Lasso and RFE kept the fewest (10 and 7, respectively). Spectral indices were retained most consistently: DBSI, NDBI, EVI, GRVI, and LSWI appeared in every subset, while HSIC-Lasso excluded NDVI and SAVI and mRMR excluded MNDWI, IBI, BAEI, NDTI, and BSI.

Hyperparameter optimization was then performed for the RF, XGBoost, and LightGBM algorithms on each selected feature subset; the resulting parameter values are summarized in Table 6. RF produced the most stable hyperparameter configurations across feature selection methods: although Boruta yielded the highest number of estimators (610), the remaining methods fell within a narrower range (436–574). In contrast, LightGBM and XGBoost showed wider variation across feature subsets, particularly in the number of estimators (LightGBM: 403–700; XGBoost: 477–689) and learning rates (LightGBM: 0.008–0.087; XGBoost: 0.012–0.043).

The classification performance obtained by the RF, XGBoost, and LightGBM models trained on each feature subset is summarized in Figure 6 as a heatmap. Among the feature selection methods, HSIC-Lasso yielded the highest and most stable

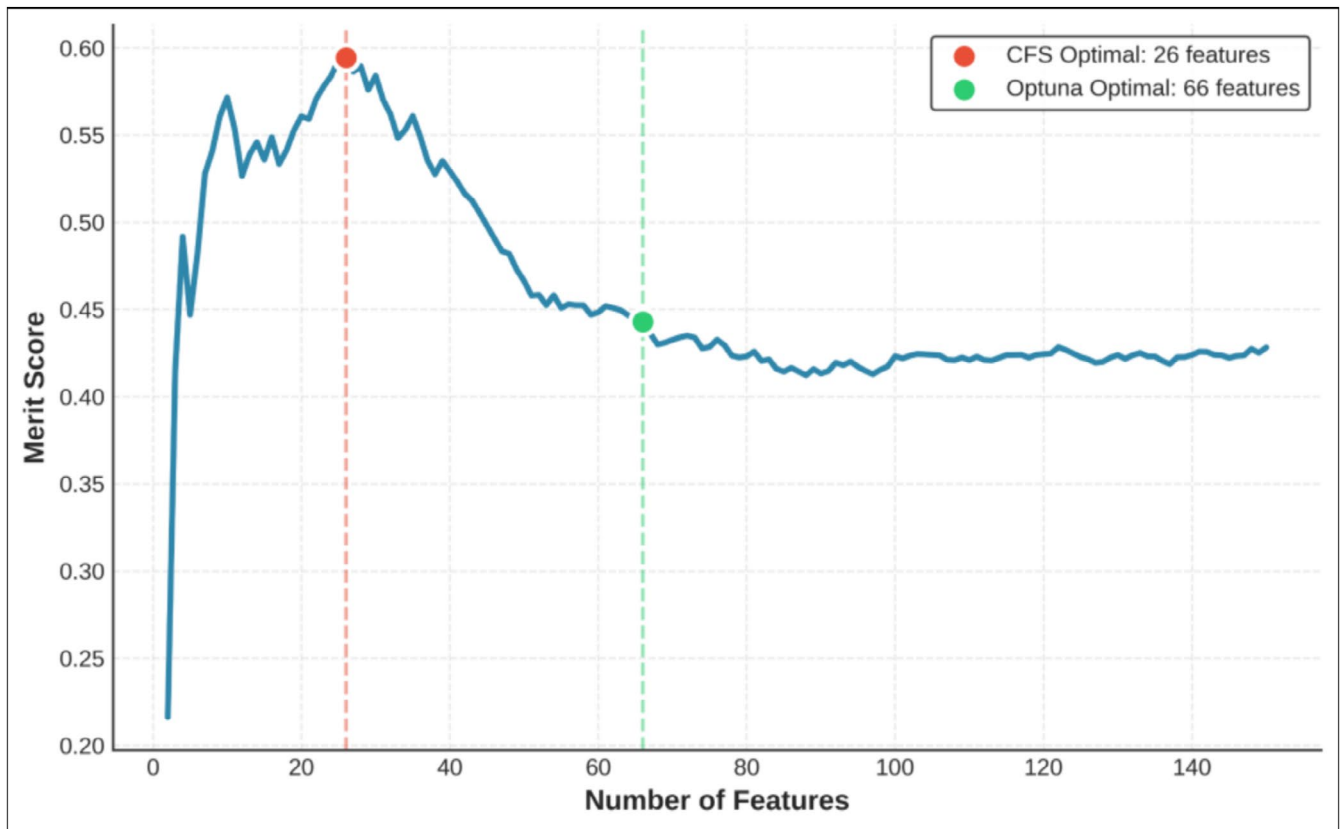


FIGURE 4 | CFS merit score curve for the ReliefF feature selection.

performance across all models, with F1 between 0.9631 and 0.9655, MD between 0.0343 and 0.0367, FA between 0.0057 and 0.0061, and SP between 0.9939 and 0.9943. McNemar's test with Holm-Bonferroni correction across 105 pairwise comparisons showed that the three HSIC-Lasso configurations were statistically equivalent to one another (RF vs. XGBoost, RF vs. LightGBM, XGBoost vs. LightGBM: adjusted $p = 1.000$). HSIC-Lasso-based configurations significantly outperformed mRMR + LightGBM and mRMR + XGBoost (e.g., HSIC-Lasso + LightGBM vs. mRMR + LightGBM: adjusted $p < 0.001$; HSIC-Lasso + RF vs. mRMR + XGBoost: adjusted $p = 0.013$), while differences with mRMR + RF did not reach statistical significance after correction (HSIC-Lasso + LightGBM vs. mRMR + RF: adjusted $p = 0.373$). HSIC-Lasso-based configurations also significantly outperformed ReliefF + RF (HSIC-Lasso + LightGBM vs. ReliefF + RF: adjusted $p = 0.002$; HSIC-Lasso + RF vs. ReliefF + RF: adjusted $p = 0.004$; HSIC-Lasso + XGBoost vs. ReliefF + RF: adjusted $p = 0.015$). In contrast, HSIC-Lasso configurations were statistically equivalent to Boruta + RF and Boruta + LightGBM (HSIC-Lasso + RF vs. Boruta + RF: adjusted $p = 1.000$; HSIC-Lasso + LightGBM vs. Boruta + RF: adjusted $p = 1.000$; HSIC-Lasso + LightGBM vs. Boruta + LightGBM: adjusted $p = 0.701$). The only exception to this equivalence was HSIC-Lasso + LightGBM versus Boruta + XGBoost (adjusted $p = 0.019$), despite their substantially different feature subset sizes (31 vs. 196). Boruta achieved similarly high performance, with F1 between 0.9552 and 0.9647; the highest F1 (0.9647) was obtained with RF, marginally below the F1 of HSIC-Lasso + LightGBM (0.9655, $\Delta F1 = 0.0008$). FA rates were slightly higher than those of HSIC-Lasso for the LightGBM and XGBoost classifiers and equal for RF, and MD ranged from

0.0348 with RF to 0.0448 with XGBoost. RFE gave stable results (F1 = 0.9546–0.9581) with moderate FA and MD values. mRMR (F1 = 0.9474–0.9543) and ReliefF (F1 = 0.9452–0.9538) yielded lower values with higher FA and MD rates; the lowest F1 was observed for ReliefF + RF (0.9452, MD = 0.0543). SP values remained consistently high across models (0.9910–0.9943). At the model level, XGBoost showed the most stable performance across feature selection methods (F1 range = 0.013), followed by LightGBM (range = 0.018) and RF (range = 0.020). The largest single drops relative to the best F1 (0.9655, HSIC-Lasso + LightGBM) were observed for ReliefF + RF ($\Delta F1 = -0.020$) and mRMR + LightGBM ($\Delta F1 = -0.018$).

To assess computational cost, the runtime of each full pipeline (feature selection + training + prediction) was measured under Google Colab (AMD EPYC 9B14, 44 cores, 172.9GB RAM, no GPU, Python 3.12.13), with prediction times averaged over 10 repetitions. All analyses were implemented in Python using scikit-learn (RF, RFE), xgboost, lightgbm, BorutaPy, pyHSIC-Lasso, mrmr-selection, skrebate (ReliefF), optuna, shap, scikit-image (GLCM), and rasterio, with Sentinel preprocessing performed in GEE. The reported durations reflect deployment costs with pre-optimized hyperparameters, while the full Optuna search (100 trials $\times$ five-fold spatial cross-validation per method) required approximately 14h. The filter-based methods were the fastest: ReliefF and mRMR completed the full pipeline in 5.7 and 6.2s on average, respectively, but achieved lower classification accuracy (mean F1 = 0.9509 and 0.9505). HSIC-Lasso required 16.6s on average while reaching the highest mean F1 (0.9645) with only 31 features. The wrapper-based methods were slower: RFE averaged 138.8s (39 features, mean F1 = 0.9568)

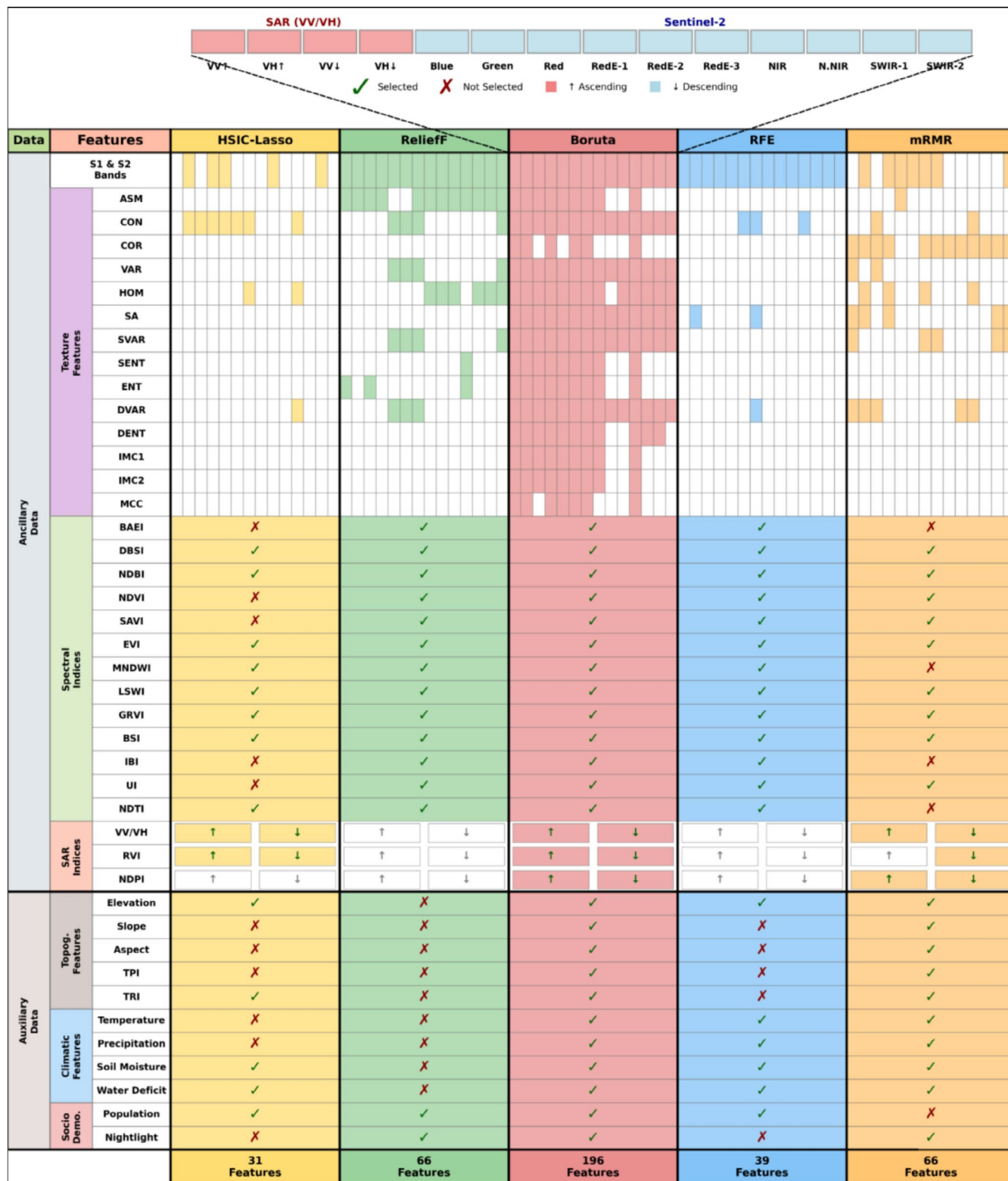


FIGURE 5 | Comparative matrix showing feature selection preferences and feature distributions across different algorithms.

and Boruta 270.4s (196 features, mean F1=0.9598). However, the one-time GLCM extraction step was completed in approximately 79 min for all 14 bands across the study area (~5.8 million pixels) using 42 parallel CPU workers, taking substantially longer than any feature selection method and thus constituting the dominant computational cost of the workflow.

Class-based prediction results are presented as confusion matrices in Figure 7. The main diagonal dominates each matrix, indicating that most samples were correctly classified. Per-class F1 scores ranged from 0.813 (bare land, ReliefF + RF) to 1.000 (water, several configurations). Bare land produced the lowest scores in every configuration, while water, forest, and cultivated

TABLE 6 | Optimized hyperparameters of RF, XGBoost, and LightGBM algorithms based on selected feature subsets.

Algorithm	Parameters	Boruta	HSIC-Lasso	mRMR	ReliefF	RFE
LightGBM	n_estimators	592	403	677	548	700
	max_depth	10	11	10	11	10
	learning_rate	0.008	0.043	0.033	0.087	0.015
	subsample	0.784	0.783	0.647	0.606	0.770
RF	n_estimators	610	436	440	455	574
	max_depth	12	13	14	17	15
	min_samples_split	6	3	10	8	10
	min_samples_leaf	2	3	2	3	5
XGBoost	n_estimators	477	618	495	563	689
	max_depth	12	10	12	11	13
	learning_rate	0.031	0.021	0.025	0.043	0.012
	subsample	0.876	0.566	0.524	0.699	0.794

Note: Search ranges: RF: n_estimators [200–800], max_depth [5–20], min_samples_split [2–10], min_samples_leaf [2–10]; XGBoost: n_estimators [200–800], max_depth [3–15], learning_rate [0.005–0.1], subsample [0.5–0.9]; LightGBM: n_estimators [200–800], max_depth [3–15], learning_rate [0.005–0.1], subsample [0.5–0.9].

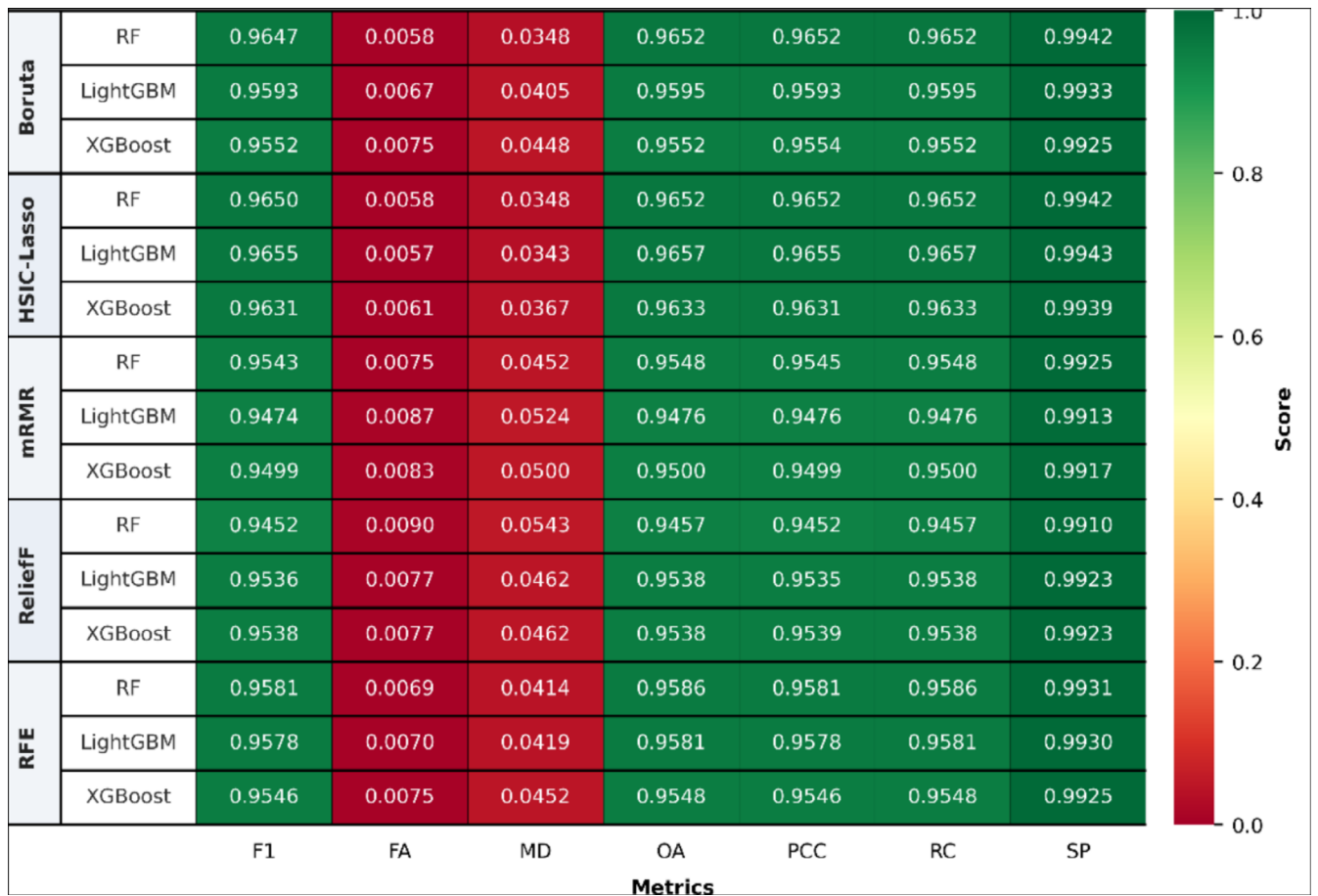


FIGURE 6 | Performance heatmap for the RF, XGBoost, and LightGBM models with optimized features and hyperparameters.

reached near-perfect classification in every classifier. For the best-performing configuration (HSIC-Lasso + LightGBM), per-class F1 values were 0.998 for water, 0.995 for forest, 0.993 for cultivated, 0.987 for urban, 0.951 for fallowland, 0.947 for road, and 0.887

for bare land. Classification errors were concentrated in specific class pairs, predominantly involving bare land in confusion with fallowland and road, and to a lesser extent with urban, across all configurations. Among the feature selection methods, HSIC-Lasso

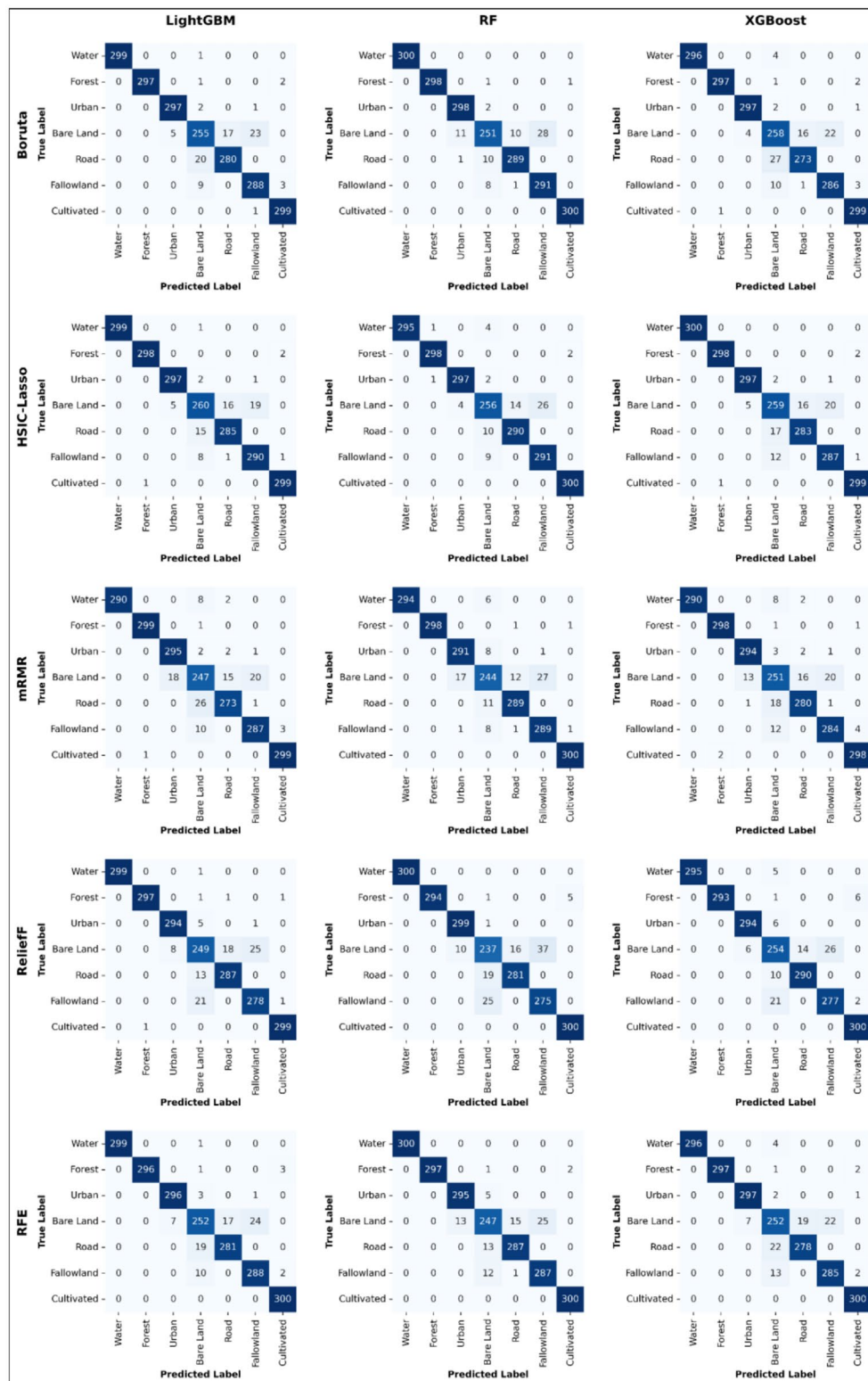


FIGURE 7 | Confusion matrices for different feature selection and model combinations.

yielded confusion matrices with the lowest off-diagonal misclassification in two classifiers (LightGBM, XGBoost) and tied with Boruta in RF. The RF+HSIC-Lasso, RF+Boruta, and LightGBM + HSIC-Lasso combinations showed the least interclass confusion. Boruta also produced high class-wise accuracy, although bare land to fallowland and road to bare land errors occurred more

frequently than in HSIC-Lasso-based models, especially with Boruta + XGBoost (road misclassified as bare land) and Boruta + RF (bare land misclassified as fallowland). RFE showed intermediate performance, with error distributions similar to Boruta but with slightly higher misclassification rates. mRMR increased misclassification across configurations, with a marked drop in bare

land accuracy and higher confusion toward road and fallowland. The LightGBM + mRMR and RF+ReliefF combinations produced the highest overall interclass confusion. ReliefF preserved high accuracy for water and forest but exhibited increased confusion among bare land, road, and fallowland, most prominently in the RF+ReliefF configuration. At the model level, XGBoost maintained clear class separation with HSIC-Lasso and Boruta but showed increased misclassification only with the mRMR subset; ReliefF and RFE remained comparable to Boruta. To examine why bare land, fallowland, and road remained the most frequently confused classes despite the multi-source design, pairwise Cohen's d was computed across feature groups. No single group separated all three. Spectral indices produced the largest effect sizes, well above Cohen's threshold for large effects ($d > 0.8$): DBSI and BSI for bare land vs. road ($d > 1$), NDBI for bare land vs. fallowland ($d > 1$), and GRVI for road vs. fallowland ($d > 1$). Elevation separated bare land (~1189m) from road and fallowland (~1000m) but did not distinguish road from fallowland ($d = 0.02$). SAR backscatter converged across the three classes, with road and fallowland nearly indistinguishable in ascending VV polarization ($d = 0.04$). GLCM textures separated road but not bare land from fallowland, and socioeconomic features produced moderate-to-large effects without strong separation (maximum $d = 0.63$). Climatic features produced moderate-to-large effect sizes for some pairs but varied primarily

with geographic location due to their 4 km native resolution rather than with land cover.

SHAP analysis was performed on the HSIC-Lasso + LightGBM configuration (Figure 8). GRVI and EVI produced the highest feature importance, followed by B11, DBSI, and elevation. GRVI contributed broadly across fallowland, urban, road, and cultivated classes, whereas EVI carried the largest weight for vegetation-related classes (forest, cultivated). B11 was most informative for distinguishing water and urban, DBSI for separating cultivated and bare land, and elevation for bare land and the agricultural classes (fallowland, cultivated). Feature importance was distributed across spectral indices, spectral bands, topographic, climatic, and SAR features, while the SAR feature VH ASC ranked lowest among the 10 most important features.

Figure 9 presents the LULC maps for all 15 configurations. Overall spatial distribution patterns are largely consistent across methods. Water is captured along the main water body in the northern part of the study area, and forest appears as broad, continuous patches, particularly in the northwestern regions. Urban forms a core cluster in the central part, while the agricultural classes (cultivated and fallowland) form

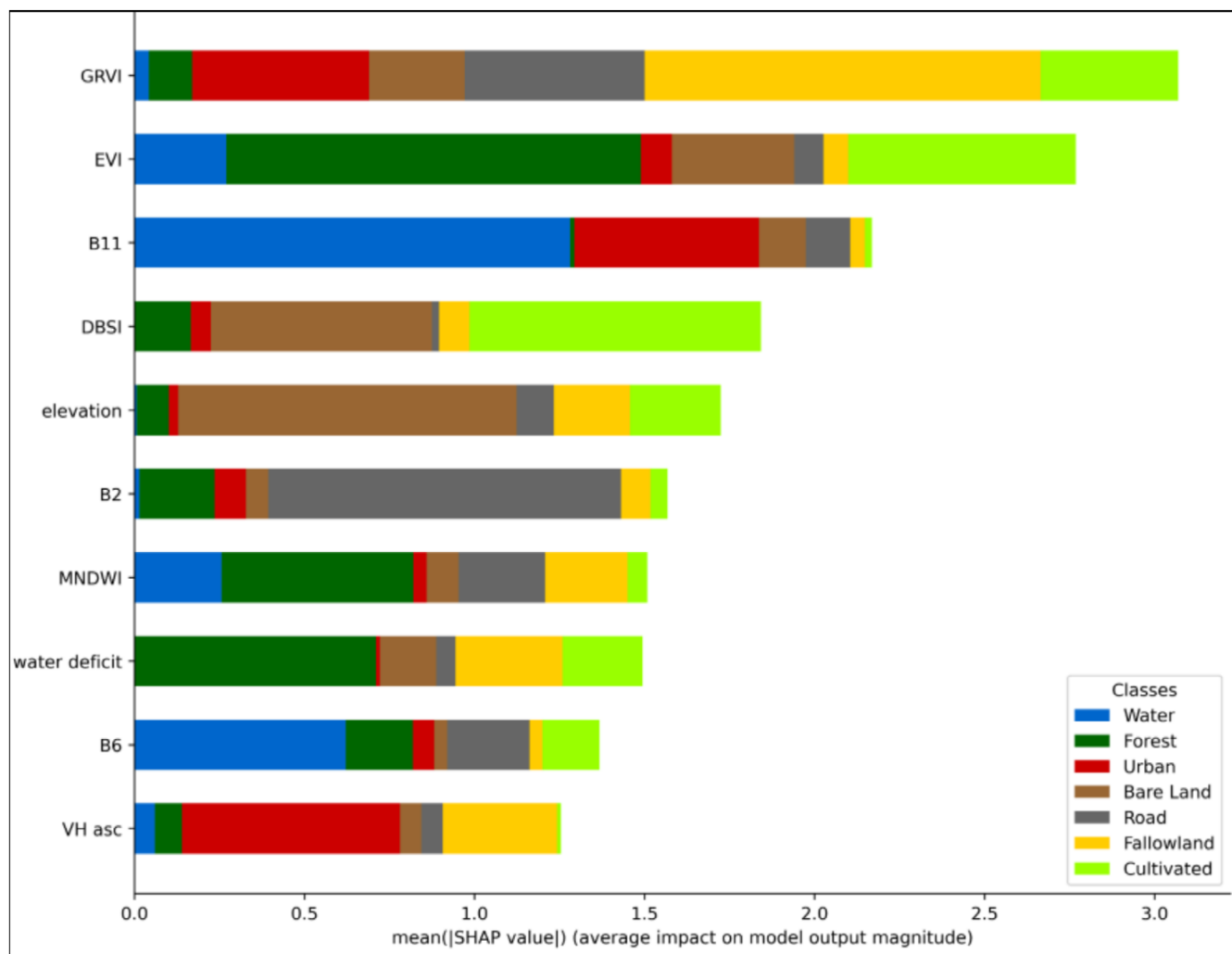


FIGURE 8 | Class-stacked SHAP summary plot for the HSIC-Lasso + LightGBM model.

a mosaic pattern mainly in the southern and eastern parts. Despite this general consistency, differences between methods appear mainly in boundary sharpness and local spatial heterogeneity. HSIC-Lasso-based maps display more uniform class regions, with compact urban areas and continuous forest and agricultural patches. ReliefF results exhibit the most salt-and-pepper effects, followed by mRMR, especially at class

boundaries. At the model level, XGBoost and LightGBM generally produced more cohesive class regions, whereas RF outputs occasionally appear more fragmented. At the class level, water and forest showed the highest spatial stability, while bare land and fallowland exhibited greater spatial variability. These spatial patterns are consistent with the class-level confusion patterns shown in Figure 7.

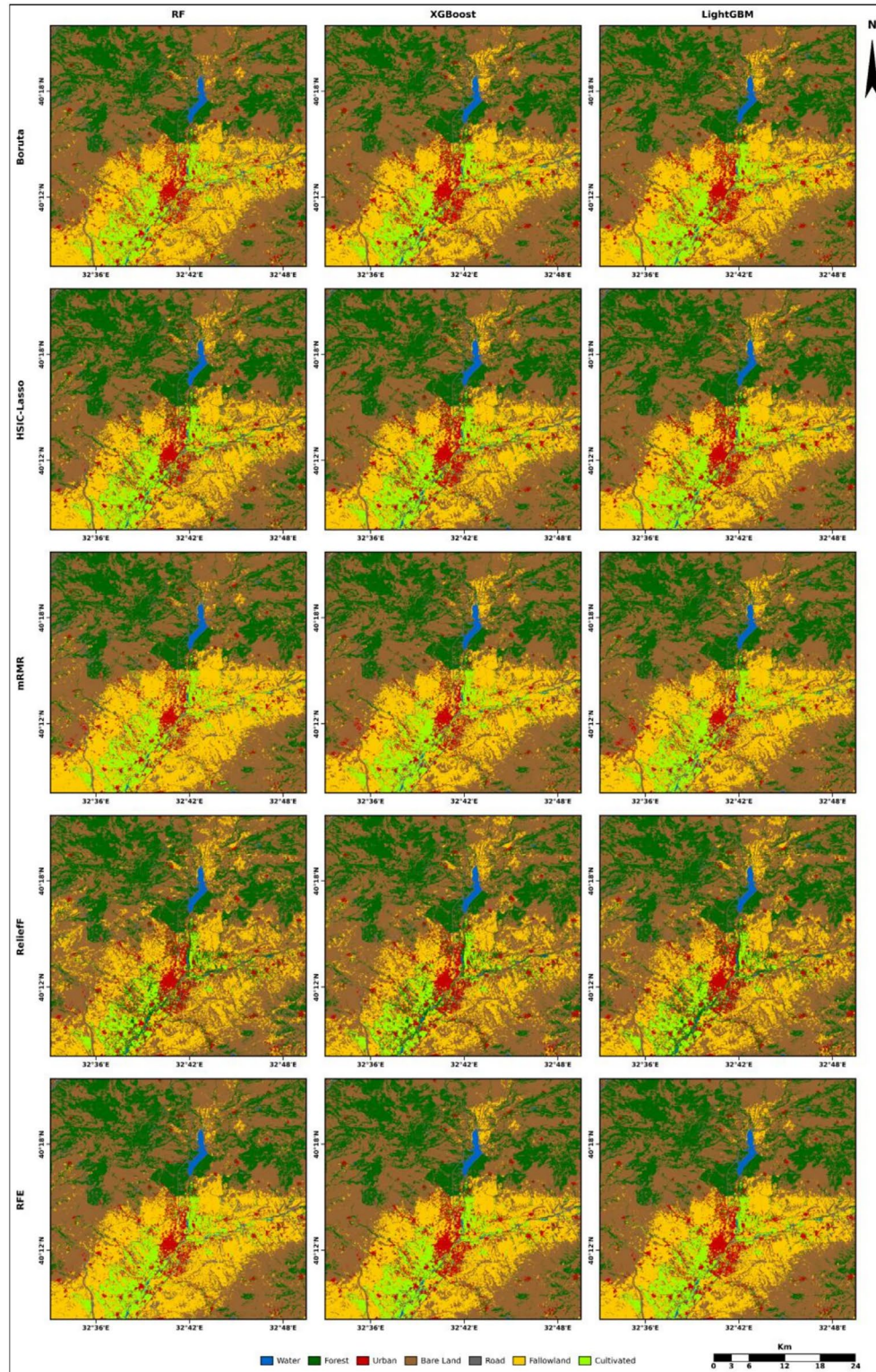


FIGURE 9 | Comparative LULC maps produced by different feature selection and model combinations.

4 | Discussion

This study evaluated LULC classification in the Kahramankazan district, northwestern Ankara, by integrating multi-source RS data with feature selection and ML techniques. The HSIC-Lasso + LightGBM configuration achieved the highest classification performance using only 31 features, 84% fewer than the 196 features retained by Boruta + RF. These results corroborate those of Kavzoglu and Bilucan (2023), who similarly found HSIC-Lasso effective in LULC classification (OA = 91.05%).

The kernel-based formulation of HSIC-Lasso captures nonlinear dependencies among features from optical, SAR, and topographic sources (Yamada et al. 2014). Despite operating on a substantially smaller subset, HSIC-Lasso preserved classification performance comparable to Boruta's much larger feature set. This contrast reflects a difference in design: Boruta retains all features that exceed shadow-feature importance, whereas HSIC-Lasso's sparsity penalty actively suppresses redundant features. In a predictor set where most features are GLCM-derived and highly correlated, this difference accounts for the reduced subset produced by HSIC-Lasso. However, this outcome does not generalize across all contexts; the suitability of a selection method depends largely on the structure of the data, as illustrated by Georganos et al. (2018), who reported that RF (the base classifier of Boruta) was largely insensitive to redundant features in a comparable very-high-resolution object-based urban classification.

At the algorithm level, the three HSIC-Lasso configurations performed within a narrow F1 range of 0.0024 and were statistically equivalent (McNemar's test with Holm-Bonferroni correction, adjusted $p = 1.000$ in all three pairwise comparisons), with LightGBM yielding the marginally highest F1 (0.9655). McCarty et al. (2020) found that LightGBM outperformed RF in large-scale LULC mapping, recording an OA of 0.653 compared with 0.594 for RF, with approximately 30% shorter processing time (287s compared with 410s). Similarly, Ustuner and Simsek (2025) reported that in an LULC classification of the Bafra Plain using Sentinel-2 data, LightGBM reached the highest accuracy among four classifiers (OA = 89.93%), exceeding RF (86.49%), Kernel Extreme Learning Machine (78.38%), and Support Vector Machine (72.49%). Abdi (2020) documented a similar pattern in boreal landscape classification with multi-temporal Sentinel-2 imagery, where XGBoost attained an OA of 0.751 compared with 0.739 for RF. These results indicate that the embedded sparsity penalty in HSIC-Lasso identifies discriminative features as effectively as Boruta's exhaustive shadow-feature comparison; the relevance-redundancy heuristic of mRMR and ReliefF tends to miss informative interactions in this feature space, with mRMR + RF as a partial exception. Although statistically significant in some pairwise comparisons, the absolute F1 differences across configurations remained small. The F1 range across feature selection methods (within a fixed classifier) reached up to 0.020, whereas the range across classifiers (within HSIC-Lasso) was only 0.0024, suggesting that feature selection exerted a stronger influence on classification performance than classifier choice when the classifiers were well-tuned. HSIC-Lasso also completed the full pipeline in 16.6s on average (16.3 times faster than Boruta), with comparable or higher F1 across the three classifiers.

Multi-source integration consistently improves classification accuracy; for example, Steinhausen et al. (2018) found that combining Sentinel-2 with Sentinel-1 imagery in the monsoon-influenced Chennai Basin increased OA from 85.85% (optical-only) to 91.53%. Zeferino et al. (2020) reported a 3-percentage-point gain (from 91% to 94%) when geological, topographic, climatic, and pedological variables were added to Landsat-8 spectral data under neotropical ecotone conditions. In a methodologically related study, Amoakoh et al. (2021) integrated Sentinel-2, Sentinel-1, and SRTM data with RFE-based feature selection in a tropical peatland landscape, achieving an OA of 94% and identifying elevation along with Sentinel-1-derived features (VH, VV, and the standard deviation of VH) as the most informative predictors.

SHAP analysis identified GRVI, EVI, B11, DBSI, and elevation as the most influential features. GRVI has been shown to outperform NDVI in distinguishing green vegetation from other land cover types (Motohka et al. 2010), and its sensitivity to vegetation presence is consistent with its broad contribution across fallowland, urban, road, and cultivated in the SHAP analysis. EVI reduces canopy-soil background effects in open canopies while remaining sensitive in dense vegetation (Huete et al. 2002), a property consistent with the high SHAP weight observed for forest and cultivated. HSIC-Lasso excluded NDVI during feature selection, and EVI emerged as the dominant vegetation index in the retained subset. The contribution of B11 to water class discrimination supports Jiang et al. (2021), who established the effectiveness of SWIR for water body detection. DBSI was originally developed for arid environments and performs well for bare land mapping under semi-arid conditions (Rasul et al. 2018); the SHAP results align with this reported sensitivity, showing high contribution to cultivated and bare land discrimination. Ibrahim (2023) similarly ranked elevation as the most important contributing feature when integrating DEM with Sentinel-1 and Sentinel-2 imagery, where this combination increased OA from 84.2% to 89.1%. Among the socioeconomic auxiliary layers, only population was retained by HSIC-Lasso; nightlight was excluded, likely because it carries a similar human-activity signal already captured by population, and the sparsity penalty in HSIC-Lasso suppresses such redundant predictors. Although population offers a strong physical contrast for urban areas, its SHAP contribution was minor compared to GRVI and B11, indicating that the spectral features carried most of the urban signal even when socioeconomic predictors were available. Spectral and topographic features have likewise dominated SHAP-based LULC interpretations elsewhere: Temenos et al. (2023) found the red, green, NIR, and SWIR bands most influential in a deep learning framework, and Chen et al. (2023) ranked elevation, NDVI, B11, and NDWI as the strongest predictors in an RF-GEE workflow. These rankings reflect model-level contributions rather than direct physical causation.

At the class level, the confusion matrices indicated that bare land, fallowland, and road produced the lowest correct prediction rates. Bare land is known to be spectrally difficult to separate from other cover types outside the growing season (Delaney et al. 2025; Nguyen et al. 2021), with Ali and Johnson (2022) reporting bare land producer accuracy as low as 87.3% in semi-arid LULC classification and up to 5.5% of pixels misclassified as fallowland. The pairwise Cohen's d analysis shows that no single feature group

resolves all three pairs: spectral indices discriminate each pair through different indices, topography isolates bare land but not road from fallowland, and SAR backscatter converges across all three for dry summer surfaces with similar roughness and low dielectric values. The residual confusion in the best-performing model indicates that combining these sources did not fully resolve this ambiguity under a single-season composite. Because multi-temporal imagery and parcel geometry were used to reduce labeling ambiguity, the observed confusion likely reflects this distributional overlap rather than labeling errors. Notably, the features that produced the largest effect sizes in this analysis (GRVI, DBSI, and elevation) also ranked among the top SHAP contributors, indicating that model importance and class separability point to the same underlying signal.

The greater salt-and-pepper noise observed in the ReliefF and mRMR maps (Figure 9) suggests that filter-based subsets, by retaining redundant features, propagate more pixel-level noise into the final classification, whereas the compact embedded subset produced by HSIC-Lasso yields spatially smoother outputs. These spatial differences are based on visual inspection of the classification maps; a quantitative spatial smoothness analysis was not performed. Comparable mRMR-based LULC studies have reached lower accuracies under similar conditions, with Wei et al. (2023) obtaining an OA of 91.37% using mRMR-based RF classification of GF-6 imagery in an urban setting. Although a direct comparison across different datasets is not possible, the higher accuracy obtained here with HSIC-Lasso (OA = 0.9657) reflects its capacity to capture nonlinear feature-class dependencies in mixed SAR-optical inputs. The mutual-information formulation of mRMR does not capture such dependencies as effectively. GLCM features contributed to capturing spatial heterogeneity, but spectral indices and topographic features were more discriminative than texture in this landscape, in line with Wu et al. (2023), who reported the same pattern in the semi-arid grasslands of northern China. Of the 240 input features, 196 were GLCM-derived, yet the best-performing subsets retained only 10 (HSIC-Lasso) and 7 (RFE) GLCM features. This disproportion, combined with the high preprocessing cost of GLCM extraction, suggests that exhaustive textural extraction may not be justified in similar semi-arid landscapes. Targeted texture computation on a reduced band subset could lower this cost without sacrificing accuracy.

These findings should be considered alongside several methodological limitations. First, the study relies on a single-season composite (June–September), which does not capture phenological changes throughout the year and contributes to the fallowland-bare land confusion discussed above. Incorporating multi-temporal composites spanning spring green-up or post-harvest periods would capture phenological signatures and improve discrimination among these classes, as reported by Sharma et al. (2018) and Andrade et al. (2021), who found notable accuracy gains over single-season composites. Second, the 10–20m spatial resolution of Sentinel-2 produces mixed-pixel effects in urban–rural transition zones, where narrower roads contribute to road-bare-land misclassification. Third, the auxiliary climate data were resampled from a 4km native resolution (TerraClimate) to 10m, limiting their spatial discriminative capacity within the 581 km² study area. Fourth, despite multi-temporal interpretation and parcel-geometry checks, the absence of in situ ground-truth verification remains a constraint for spectrally ambiguous classes. Two further

limitations concern the statistical evaluation: first, McNemar's test assumes independence among test samples, and although a spatial split separated the training and test regions, residual spatial autocorrelation within the test set may inflate Type I error rates; second, the balanced training and test sets used here (700 and 300 samples per class) do not reflect the natural class proportions in the landscape, so performance may differ under imbalanced real-world distributions.

5 | Conclusion

HSIC-Lasso + LightGBM achieved the best performance among the 15 evaluated configurations, using the most compact feature subset (31 features); compact embedded selection thus emerges as a competitive alternative to larger wrapper-based subsets in heterogeneous peri-urban multi-source LULC workflows. The choice of feature selection method appeared to have a stronger influence on classification accuracy than the choice of classifier, with F1 ranging up to 0.020 across feature selection methods compared with 0.0024 across the three HSIC-Lasso configurations. The three HSIC-Lasso configurations were statistically indistinguishable from one another and remained equivalent to the best-performing Boruta variants, while significantly outperforming the mRMR-based and ReliefF + RF baselines (Holm-Bonferroni-adjusted $p < 0.05$). The resulting maps can support land management and peri-urban monitoring in the district, and the computational efficiency of HSIC-Lasso makes regular map production feasible at district or provincial scales.

Future work could extend this comparative approach in three directions. Multi-temporal image composites incorporating spring and post-harvest phenological windows would improve discrimination between spectrally similar classes, such as fallowland and bare land. Object-based classification approaches and higher-resolution imagery can reduce salt-and-pepper noise and mixed-pixel effects observed in pixel-based outputs, particularly in urban–rural transition zones. Applying this comparative approach to other study areas would clarify whether the performance patterns observed here are specific to Kahramankazan or recur across different settings.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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